1.2 Infinite Sequences

1.2.1 Definition of a Sequence

Definition 1.2.1

An **infinite sequence** (ลำดับอนันต์) is a function whose domain is the set of integers.

Typically, the domain of a sequence is the set of *positive integers* or the set of *nonnegative integers*. We will regard the expression $a_1, a_2, a_3, \ldots, a_n, \ldots$ to be an alternative notation for the function $f(n) = a_n,\ n = 1, 2, 3, \ldots$, and we will regard $a_0, a_1, a_2, \ldots, a_n, \ldots$ to be an alternative notation for the function $f(n) = a_n,\ n = 0, 1, 2, \ldots$.

When the general term of a sequence

$$a_1, a_2, a_3, \ldots, a_n, \ldots \tag{1.6}$$

is known, there is no need to write out the initial terms, and it is common to write only the general term enclosed in braces. Thus, (1.6) might be written as

$$\{a_n\}_{n=1}^{+\infty} \quad \text{or} \quad \{a_n\}_{n=1}^{\infty}$$

For example, the sequence $\frac{1}{2}, \frac{2}{3}, \frac{3}{4}, \ldots, \frac{n}{n+1}, \ldots$ can be written as $\left\{\frac{n}{n+1}\right\}_{n=1}^{+\infty}$, or equivalently $\left\{\frac{n+1}{n+2}\right\}_{n=0}^{+\infty}$

1.2.2 Graphs of Sequences

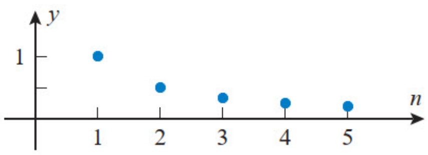
Since sequences are functions, it makes sense to talk about the graph of a sequence. For example, the graph of the sequence $\left\{\frac{1}{n}\right\}_{n=1}^{+\infty}$ is the graph of the equation

$$y = \frac{1}{n}, \quad n = 1, 2, 3, \ldots$$

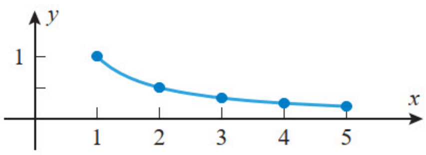
Because the right side of this equation is defined only for positive integer values of n, the graph consists of a succession of isolated points. This is different from the graph of

$$y = \frac{1}{x}, \quad x \geq 1$$

which is a continuous curve.



$y = \frac{1}{n},\ n = 1, 2, 3, \ldots$

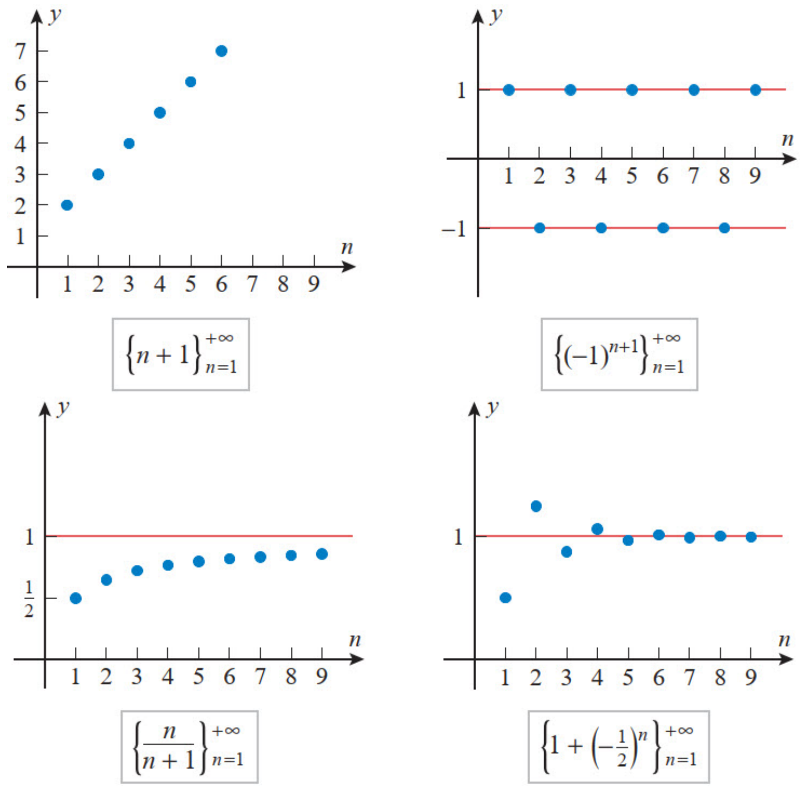


$y = \frac{1}{x},\ x \geq 1$

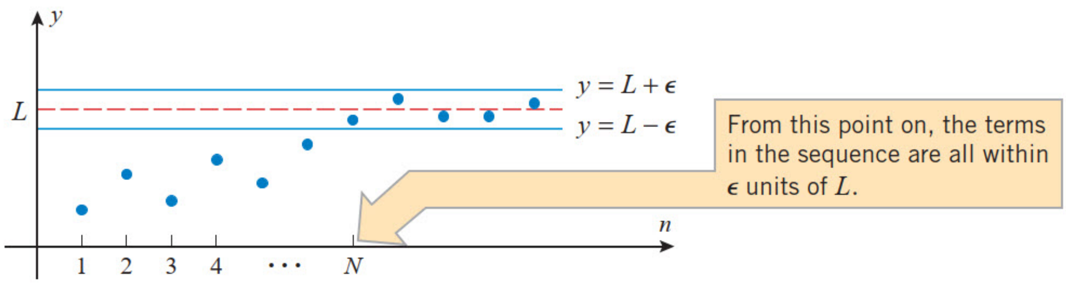
1.2.3 Limit of a Sequence

Since sequences are functions, we can inquire about their limits. However, because a sequence $\{a_n\}_{n=1}^{+\infty}$ is only defined for integer values of n, the only limit that makes sense is the limit of a_n as $n \to +\infty$. In the below figures, we have shown the graphs of four sequences, each of which behaves differently as

- The terms in the sequence $\{n+1\}$ increase without bound.
- The terms in the sequence $\{(-1)^{n+1}\}$ oscillate between -1 and 1.
- The terms in the sequence $\left\{\dfrac{n}{n+1}\right\}$ increase toward a “limiting value” of 1.
- The terms in the sequence $\left\{1+\left(-\dfrac{1}{2}\right)^n\right\}$ also tend toward a “limiting value” of 1, but do so in an oscillatory fashion.



Informally speaking, the limit of a sequence $\{a_n\}$ is intended to describe how an behaves as $n \to +\infty$. To be more specific, we will say that a *sequence* $\{a_n\}$ *approaches a limit* L *if the terms in the sequence eventually become arbitrarily close to* L. Geometrically, this means that for any positive number ε there is a point in the sequence after which all terms lie between the lines $y = L - \varepsilon$ and $y = L + \varepsilon$.



The following definition makes these ideas precise.

Definition 1.2.2

A sequence $\{a_n\}$ is said to **converge** (ลู่เข้า) to the **limit** (ลิมิต) L if given any $\varepsilon > 0$, there is a positive integer N such that $|a_n - L| < \varepsilon$ for $n \geq N$. In this case we write

$$\lim_{n\to+\infty} a_n = L$$

A sequence that does not converge to some finite limit is said to **diverge** (ลู่ออก).

Example 1.2.3

- The sequences $\left\{\frac{n}{n+1}\right\}$ and $\left\{1+\left(-\frac{1}{2}\right)^n\right\}$ are convergent and

$$\lim_{n\to+\infty} \frac{n}{n+1} = 1 \quad \text{and} \quad \lim_{n\to+\infty}\left(1+\left(-\frac{1}{2}\right)^n\right) = 1.$$

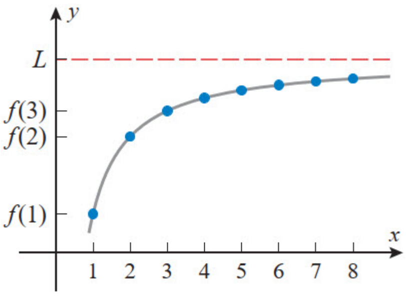
- The sequences $\{n+1\}$ and $\left\{(-1)^{n+1}\right\}$ are divergent.

Theorem 1.2.4

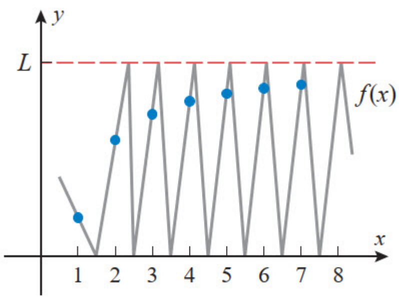
If $\lim\limits_{n\to+\infty} f(x) = L$ and $f(n) = a_n$ when n is an integer, then $\lim\limits_{n\to+\infty} a_n = L$.

In particular, since we know that $\lim\limits_{n\to+\infty} \frac{1}{x^r} = 0$ when $r > 0$, we have

$$\lim_{n\to+\infty} \frac{1}{n^r} = 0 \text{ if } r > 0. \tag{1.7}$$



If $f(x) \to L$ as $x \to +\infty$, then $f(n) \to L$ as $n \to +\infty$.



$f(n) \to L$ as $n \to +\infty$, but $f(x)$ diverges by oscillation as $x \to +\infty$.

Remark 1.2.5

The converse of Theorem 1.2.4 is not true; that is, one *cannot* infer that $f(x) \to L$ as $x \to +\infty$ from the fact that $f(n) \to L$ as $n \to +\infty$.

If a_n becomes large as n becomes large, we use the notation $\lim\limits_{n\to+\infty} a_n = +\infty$.

Definition 1.2.6

$\lim_{n\to+\infty} a_n = +\infty$ means that for every positive number M there is an integer N such that

$$\text{if } \ n > N, \ \text{ then } \ a_n > M.$$

If $\lim_{n\to+\infty} a_n = +\infty$, then $\{a_n\}$ is divergent but in a special way. We say that $\{a_n\}$ diverges to $+\infty$. We define $\lim_{n\to+\infty} a_n = -\infty$ similarly.

Example 1.2.7: Geometric Sequences with $r > 0$

Prove that for $r > 0$ and $c > 0$

$$\lim_{n\to+\infty} cr^n = \begin{cases} 0 & \text{if } \ 0 < r < 1 \\ c & \text{if } \ r = 1 \\ +\infty & \text{if } \ r > 1 \end{cases}$$

Solution. Set $f(x) = cr^x$. If $0 < r < 1$, then, by Theorem 1.2.4,

$$\lim_{n\to+\infty} cr^n = \lim_{x\to+\infty} f(x) = c \lim_{x\to+\infty} r^x = 0.$$

If $r > 1$, then, since $c > 0$, both $f(x)$ and the sequence $\{cr^n\}$ diverge to $+\infty$. If $r = 1$, then $cr^n = c$ for all n, and the limit is c. $\square$

The following theorem, which we state without proof, shows that the familiar properties of limits apply to sequences

Theorem 1.2.8

Suppose that the sequences $\{a_n\}$ and $\{b_n\}$ converge to limits L_1 and L_2, respectively, and c is a constant. Then:

1. $\lim_{n\to\infty} c = c$
2. $\lim_{n\to\infty} ca_n = c\left(\lim_{n\to\infty} a_n\right) = cL_1$
3. $\lim_{n\to\infty} (a_n \pm b_n) = \lim_{n\to\infty} a_n \pm \lim_{n\to\infty} b_n = L_1 \pm L_2$
4. $\lim_{n\to\infty} (a_n b_n) = \left(\lim_{n\to\infty} a_n\right)\left(\lim_{n\to\infty} b_n\right) = L_1 L_2$
5. $\lim_{n\to\infty} \left(\dfrac{a_n}{b_n}\right) = \dfrac{\lim_{n\to\infty} a_n}{\lim_{n\to\infty} b_n} = \dfrac{L_1}{L_2}$ (if $L_2 \neq 0$)
6. $\lim_{n\to\infty} (a_n)^p = \left(\lim_{n\to\infty} a_n\right)^p = L_1^p$ if $p > 0$ and $a_n > 0$

Example 1.2.9

Determine whether the sequence converges or diverges. If it converges, find the limit.

(a) $\left\{ \dfrac{n}{2n+1} \right\}_{n=1}^{+\infty}$

(b) $\left\{ \dfrac{3n^2}{n^3 - n + 1} \right\}_{n=1}^{+\infty}$

(c) $\left\{ \sqrt{n^2-1} - n \right\}_{n=1}^{+\infty}$

(d) $\left\{ \dfrac{n}{\sqrt{10+n}} \right\}_{n=1}^{+\infty}$

(e) $\{8 - 2n\}_{n=1}^{+\infty}$

Theorem 1.2.10

If $\lim_{n\to+\infty} a_n = L$ and the function f is continuous at L, then

$$\lim_{n\to+\infty} f(a_n) = f\left(\lim_{n\to+\infty} a_n\right) = f(L).$$

Example 1.2.11

Find $\lim_{n\to+\infty} \sin\left(\frac{\pi}{n}\right)$.

Solution. Because the sine function is continuous at 0, Theorem 1.2.10 enables us to write

$$\lim_{n\to+\infty} \sin\left(\frac{\pi}{n}\right) = \sin\left(\lim_{n\to+\infty} \frac{\pi}{n}\right) = \sin 0 = 0.$$

□

Sometimes the even-numbered and odd-numbered terms of a sequence behave sufficiently differently that it is desirable to investigate their convergence separately. The following theorem, whose proof is omitted, is helpful for that purpose.

Theorem 1.2.12

A sequence converges to a limit L if and only if the sequences of even-numbered terms and odd-numbered terms both converge to L.

Example 1.2.13

The sequence

$$\frac{1}{2}, \frac{1}{3}, \frac{1}{2^2}, \frac{1}{3^2}, \frac{1}{2^3}, \frac{1}{3^3}, \ldots$$

converges to 0, since the even-numbered terms and the odd-numbered terms both converge to 0, and the sequence

$$1, \frac{1}{2}, 1, \frac{1}{3}, 1, \frac{1}{4}, \ldots$$

diverges, since the odd-numbered terms converge to 1 and the even-numbered terms converge to 0.

1.2.4 The Squeezing Theorem for Sequences

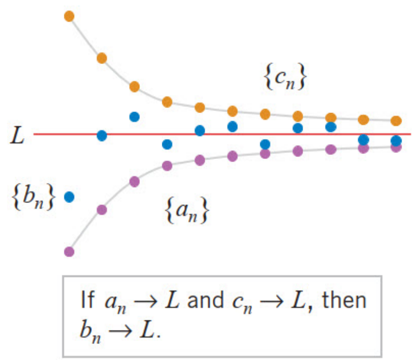
The following theorem will be useful for finding limits of sequences that cannot be obtained directly. The proof is omitted.

Theorem 1.2.14: The Squeezing Theorem

Let $\{a_n\}$, $\{b_n\}$ and $\{c_n\}$ be sequences such that

$a_n \leq b_n \leq c_n$ (for all values of n beyond some index N).

If the sequences $\{a_n\}$ and $\{c_n\}$ have a common limit L as $n \to +\infty$, then $\{b_n\}$ also has the limit L as $n \to +\infty$.



Example 1.2.15

Find the limit of the sequence

$$\left\{ \frac{n!}{n^n} \right\}_{n=1}^{+\infty}$$

The following theorem is often useful for finding the limit of a sequence with both positive and negative terms.

Theorem 1.2.16

If $\lim_{n\to\infty} |a_n| = 0$, then $\lim_{n\to\infty} a_n = 0$.

Proof. Since $-|a_n| \leq a_n \leq |a_n|$ and the limit of the two outside terms is 0, we have $\lim_{n\to\infty} a_n = 0$ by the Squeezing Theorem. $\square$

Example 1.2.17

As $\left|(-1)^n \frac{1}{2^n}\right| = \frac{1}{2^n}$ and $\lim_{n\to\infty} \frac{1}{2^n} = 0$, by Theorem 1.2.16,

$$\lim_{n\to\infty} (-1)^n \frac{1}{2^n} = 0.$$

Example 1.2.18

For what values of r is the sequence $\{r^n\}$ convergent?

Solution. By Example 1.2.17, it is enough to consider only the case that $r \leq 0$. It is obvious that $\lim_{n\to+\infty} 0^n = 0$. If $-1 < r < 0$, then $0 < |r| < 1$, so , by Example 1.2.17, we have $\lim_{n\to+\infty} |r| = 0$, and hence

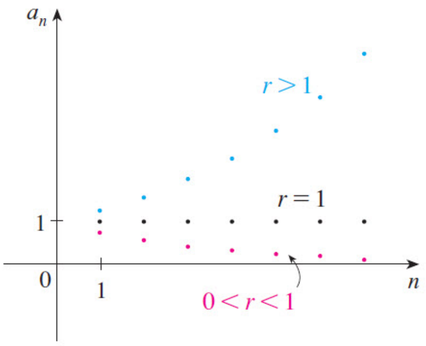
$$\lim_{n\to+\infty} |r^n| = \lim_{n\to+\infty} |r|^n = 0.$$

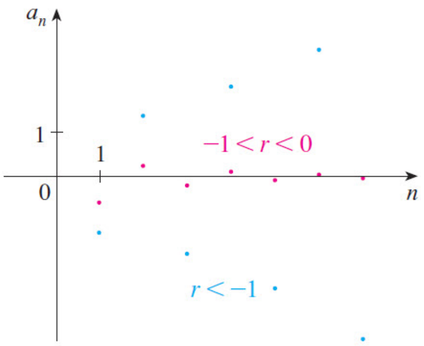
Therefore, $\lim_{n\to+\infty} r^n = 0$ by Theorem 1.2.16. If $r \leq -1$, then $\{r^n\}$ diverges as in Example 1.2.3. $\square$

The results of Example 1.2.18 are summarized for future use as follows.

The sequence $\{r^n\}$ is convergent if $-1 < r \leq 1$ and divergent for all other values of r.

$$\lim_{n\to+\infty} r^n = \begin{cases} 0 & \text{if } -1 < r < 1 \\ 1 & \text{if } r = 1 \end{cases}$$





1.2.5 Convergence of Monotone Sequences

Definition 1.2.19

A sequence $\{a_n\}$ is called **increasing** if $a_n < a_{n+1}$ for all $n \geq 1$, that is, $a_1 < a_2 < a_3 < \ldots$. It is called **decreasing** if $a_n > a_{n+1}$ for all $n \geq 1$. A sequence is **monotonic** if it is either increasing or decreasing.

Example 1.2.20

The sequence $\left\{\dfrac{3}{n+5}\right\}$ is decreasing because $\dfrac{3}{n+5} > \dfrac{3}{n+6} = \dfrac{3}{(n+1)+5}$ and so $a_n > a_{n+1}$ for all $n \geq 1$.

Definition 1.2.21

A sequence $\{a_n\}$ is **bounded above** if there is a number M such that

$$a_n \leq M \quad \text{for all } n \geq 1.$$

It is **bounded below** if there is a number m such that

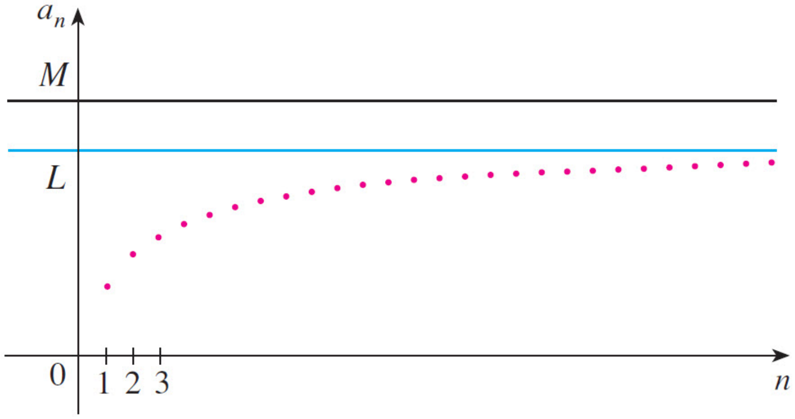
$$m \leq a_n \quad \text{for all } n \geq 1.$$

If it is bounded above and below, then $\{a_n\}$ is a **bounded sequence**.

Example 1.2.22

- The sequence $a_n = n$ is bounded below (because $1 \leq a_n$ for all $n \geq 1$) but not above.
- The sequence $a_n = \dfrac{n}{n+1}$ is bounded because $0 < a_n < 1$ for all $n \geq 1$.

We know that not every bounded sequence is convergent for instance, the sequence $a_n = (-1)^n$ satisfies $-1 \leq a_n \leq 1$ but is divergent, and not every monotonic sequence is convergent for instance, the sequence $a_n = n \to +\infty$. But if a sequence is both bounded and monotonic, then it must be convergent.



Theorem 1.2.23: Monotonic Sequence Theorem

Every bounded, monotonic sequence is convergent.

It follows from Theorem 1.2.23 that

- a sequence that is increasing and bounded above is convergent,
- a sequence that is decreasing and bounded below is convergent.

Example 1.2.24

Investigate the sequence $\{a_n\}$ defined by the *recurrence relation*

$$a_1 = 2, \quad a_{n+1} = \frac{1}{2}(a_n + 6) \quad \text{for } n = 1, 2, 3, \ldots$$

1.2.6 Exercises

1. Determine whether the sequence converges, and if so find its limit.

   1.1. $0.9, 0.99, 0.999, 0.9999, \ldots$

   1.2. $2, \left(\frac{3}{2}\right)^2, \left(\frac{4}{3}\right)^3, \left(\frac{5}{4}\right)^4, \ldots$

   1.3. $\left\{\sin\left(\frac{n\pi}{4}\right)\right\}$

   1.4. $\left\{\frac{\cos n}{n}\right\}$

   1.5. $\left\{\frac{(n+1)^n}{n^{n+1}}\right\}$

   1.6. $\left\{\frac{(n+1)\ln n - n\ln(n+1)}{\ln n}\right\}$

   1.7. $\left\{\sqrt[n]{a^n + b^n}\right\}$ if $0 < a < b$

   1.8. $\left\{\frac{1-3n}{2n+1}\right\}$

   1.9. $\{1 + (-1)^n\}$

   1.10. $\left\{\left(\frac{n+1}{2n}\right)\left(1 - \frac{1}{n}\right)\right\}$

   1.11. $\left\{\sqrt{n^2+n} - n\right\}$

   1.12. $\left\{\frac{2^{1000} + 2^{n-1} + 3^{n-2}}{2^n + 3^n + 5}\right\}$

   1.13. $\left\{\frac{1+2+3+\ldots+n}{n+1}\right\}$

   1.14. $\left\{\frac{e^n - e^{-n}}{e^n + e^{-n}}\right\}$

   1.15. $\left\{\left(1 - \frac{1}{7n}\right)^n\right\}$ *Hint:* $\lim\limits_{n\to+\infty}\left(1 + \frac{1}{n}\right)^n = e$

2. For what values of x does the sequence $\{a_n\}_{n=1}^{+\infty}$ whose nth term is $a_n = \left(\frac{2x^2+1}{x^2+5}\right)^n$ converge?

3. Let k be a constant and $\{a_n\}_{n=1}^{+\infty}$ a sequence whose nth term is

$$a_n = \frac{1}{n^k}\left[1 + (2+2) + (3+3+3) + \ldots + \overbrace{(n+n+\ldots+n)}^{n \text{ terms}}\right].$$

   If $\lim\limits_{n\to\infty} a_n = \ell$ for some $\ell > 0$, what is the value of $6(\ell + k)$?

4. Let $\{a_n\}_{n=1}^{+\infty}$ and $\{b_n\}_{n=1}^{+\infty}$ be two sequences whose nth term are

$$a_n = \frac{n(1+2+3+...+n)}{3(1^2+2^2+3^2+...+n^2)} \quad \text{and} \quad b_n = \frac{\sqrt{3n+2}-\sqrt{3n+1}}{\sqrt{n+2}-\sqrt{n+1}}.$$

Find $\lim\limits_{n\to\infty}(a_n+b_n)$.

5. Find $\lim\limits_{n\to\infty} \dfrac{\left(1^4+\frac{1}{4}\right)\left(3^4+\frac{1}{4}\right)\left(5^4+\frac{1}{4}\right)\cdots\left((2n-1)^4+\frac{1}{4}\right)}{\left(2^4+\frac{1}{4}\right)\left(4^4+\frac{1}{4}\right)\left(6^4+\frac{1}{4}\right)\cdots\left((2n)^4+\frac{1}{4}\right)}$.

Hint: Apply the Sophie Germain Identity: $a^4+4b^4=\left(a^2+2ab+2b^2\right)\left(a^2-2ab+2b^2\right)$

6. Determine whether the statement is true or false. Explain your answer.

6.1. Sequences are functions.

6.2. If $\{a_n\}$ and $\{b_n\}$ are sequences such that $\{a_n+b_n\}$ converges, then $\{a_n\}$ and $\{b_n\}$ converge.

6.3. If $\{a_n\}$ and $\{b_n\}$ are divergent sequences, then $\{a_n+b_n\}$ is divergent.

6.4. If $\{a_n\}$ and $\{b_n\}$ are divergent sequences, then $\{a_nb_n\}$ is divergent.

6.5. If $\lim\limits_{n\to+\infty} a_n = 0$ and $\lim\limits_{n\to+\infty} b_n = 0$, then $\left\{\dfrac{a_n}{b_n}\right\}$ converges.

6.6. If $\{a_n\}$ diverges, then $a_n\to+\infty$ or $a_n\to-\infty$ as $n\to+\infty$.

6.7. If the graph of $y=f(x)$ has a horizontal asymptote as $x\to+\infty$, then the sequence $\{f(n)\}$ converges.

7. Give two examples of sequences, all of whose terms are between -10 and 10, that do not converge. Use graphs of your sequences to explain their properties.

8. 8.1. Suppose that f satisfies $\lim\limits_{x\to0^+} f(x)=+\infty$. Is it possible that the sequence $\{f(1/n)\}$ converges? Explain.

8.2. Find a function f such that $\lim\limits_{x\to0^+} f(x)$ does not exist but the sequence $\{f(1/n)\}$ converges.

9. (a) Starting with $n=1$, and considering the even and odd terms separately, find a formula for the general term of the sequence

$$1, \frac{1}{3}, \frac{1}{3}, \frac{1}{5}, \frac{1}{5}, \frac{1}{7}, \frac{1}{7}, \frac{1}{9}, \frac{1}{9}, \ldots$$

(b) Determine whether the sequences in part (a) converge. For those that do, find the limit.

10. For what positive values of b does the sequence $b, 0, b^2, 0, b^3, 0, b^4, \ldots$ converge? Justify your answer.

11. Use the Squeezing Theorem to prove that

$$\lim_{n\to+\infty} \frac{R^n}{n!} = 0$$

for all real number R.

12. Consider the sequence

$$\begin{aligned} a_1 &= \sqrt{6} \\ a_2 &= \sqrt{6+\sqrt{6}} \\ a_3 &= \sqrt{6+\sqrt{6+\sqrt{6}}} \\ a_4 &= \sqrt{6+\sqrt{6+\sqrt{6+\sqrt{6}}}} \\ &\vdots \end{aligned}$$

 12.1. Find a recursion formula for a_{n+1}.

 12.2. Assuming that the sequence converges, find the limit.

13. Let $\{a_n\}_{n=1}^{+\infty}$ be the sequence defined recursively by

$$a_1 = \sqrt{2} \quad \text{and} \quad a_{n+1} = \sqrt{2 + a_n} \quad \text{for } n \geq 1.$$

 (a) List the first three terms of the sequence.

 (b) Show that $a_n < 2$ for $n \geq 1$.

 (c) Show that $a_{n+1}^2 - a_n^2 = (2 - a_n)(1 + a_n)$ for $n \geq 1$.

 (d) Use the results in parts (b) and (c) to show that $\{a_n\}_{n=1}^{+\infty}$ is an increasing sequence.

 (e) Show that $\{a_n\}_{n=1}^{+\infty}$ converges and find its limit L.

14. A patient is to take a 50-mg pill of a certain drug every morning. It is known that the body eliminates 40% of the drug every 24 h.

 14.1. Find a recursive sequence that models the amount A_n of the drug in the patient's body after each pill is taken.

 14.2. Find the first four terms of the sequence A_n and a formula for A_n.

 14.3. How much of the drug remains in the patient's body after 5 days? How much will accumulate in his system after prolonged use?

15. Let

$$f(x) = \begin{cases} 2x, & 0 \leq x < 0.5 \\ 2x - 1, & 0.5 \leq x < 1 \end{cases}$$

Does the sequence $f(0.2), f(f(0.2)), f(f(f(0.2))), ...$ converge? Justify your reasoning.

16. 16.1. Use a graphing utility to generate the graph of the equation $y = (2^x + 3^x)^{1/x}$, and then use the graph to make a conjecture about the limit of the sequence

$$\left\{(2^n + 3^n)^{1/n}\right\}_{n=1}^{+\infty}$$

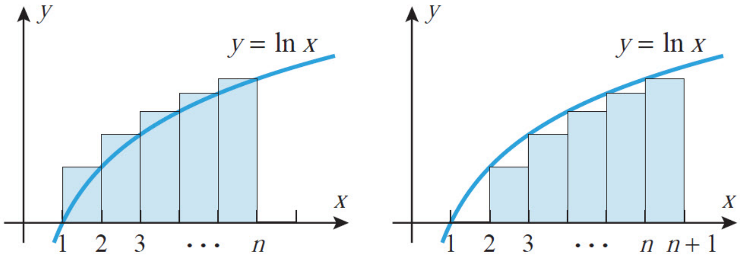
16.2. Confirm your conjecture by calculating the limit.

17. Consider the sequence $\{a_n\}_{n=1}^{+\infty}$ whose nth term is $a_n = \dfrac{1}{n}\displaystyle\sum_{k=1}^{n} \dfrac{1}{1+(k/n)}$.

Show that $\displaystyle\lim_{n\to+\infty} a_n = \ln 2$ by interpreting a_n as the Riemann sum of a definite integral.

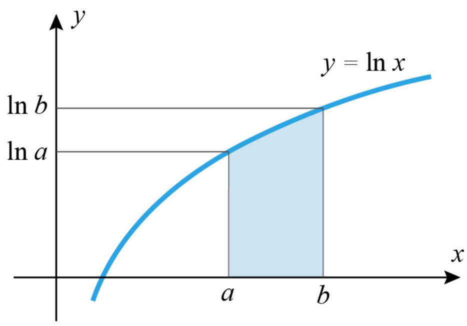
18. (a) Compare appropriate areas in the accompanying figure to deduce the following inequalities for $n \geq 2$:

$$\int_1^n \ln x\, dx < \ln n! < \int_1^{n+1} \ln x\, dx$$



(b) Use the accompanying figure to deduce that

$$\int_a^b \ln x\, dx = b\ln b - a\ln a - \int_{\ln a}^{\ln b} e^y\, dy$$



(c) Use the results in parts (a) and (b) to show that

$$\frac{n^n}{e^{n-1}} < n! < \frac{(n+1)^{n+1}}{e^n}, \quad n > 1$$

(d) Use AM-GM inequality and the Squeezing Theorem to show that $\displaystyle\lim_{n\to+\infty} (n+1)^{1/n} = 1$.

(e) Use the Squeezing Theorem and the results in parts (c) and (d) to show that

$$\lim_{n\to+\infty} \frac{\sqrt[n]{n!}}{n} = \frac{1}{e}$$

(f) Use the left inequality in part (c) to show that

$$\lim_{n\to+\infty} \sqrt[n]{n!} = +\infty$$